Block Order under a Fixed English–Tagalog Token Budget:
A One-Seed Schedule-Topology Study with nanochat

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Abstract

Holding English–Tagalog source-content token quotas, tokenizer, parent type, phase-two budget, and optimizer-reset policy fixed, does phase-two *schedule topology*—the order and granularity of language blocks—change Tagalog and English full-split validation bits-per-byte (BPB)? This question was preregistered as AsPredicted #307969 after P5 Gate W and before any P6 parent, child, validation BPB, or restricted-test BPB existed. P6-M is a post-P5, prospectively preregistered one-seed schedule-topology mechanism study. It does not amend AsPredicted #306780, #306935, #307342, #307591, or #307836. It is not a P5-style recurrence-count study. One parent-initialization seed ($s = 4$) trained a fresh Tagalog parent on WikiText-TL-39 with Karpathy's nanochat trainer pinned at commit `92d63d4`, the carry-forward P4 tokenizer, $T = 2048$, and NVIDIA CUDA A40. After P0-T PASS, d20 was frozen as immutable C0. Six matched d20 continuations compared C1 extra Tagalog, C2 pure English, and four mixed topologies (M-fine, M-coarse, M-blocked, M-rand) that each use exactly 9,633,792 Tagalog and 9,633,792 English source-content tokens for D_{phase2} = 19,267,584 tokens ($N = 294$). Primary estimands are $\Delta\text{TL}(\tau) = \text{TL}(M\text{-}\tau) - \text{TL}(\text{M-fine})$ and $\Delta\text{EN}(\tau) = \text{EN}(M\text{-}\tau) - \text{EN}(\text{M-fine})$ for $\tau \in \{\text{coarse}, \text{blocked}, \text{rand}\}$ at $\delta = 0.01$. At Gate X, M-coarse was better than M-fine on Tagalog and worse on English; M-blocked was worse on both; M-rand was within δ on both. One authorized M-fine-only secondary test is reported separately and excluded from topology classification. One seed; no confidence interval, mean, p-value, or population claim.

Keywords: schedule topology; block order; token budget; Tagalog; English; bits-per-byte; nanochat; preregistration; post-P5; one seed.

arXiv categories: cs.CL, cs.LG.

Registration and deposits (P6-M): AsPredicted #307969 (https://aspredicted.org/bk6m9d.pdf); ResearchBox #8918 (https://researchbox.org/8918; FOR PEER REVIEW, not Make Public); AsCollected #2541 Version 1 (https://ascollected.org/XZ8_TI5; project `NANOCHAT-FILIPINO P6-M`; linked from box #8918); study record https://github.com/pageman/nanochat-filipino/tree/main/docs/p6; GitHub trees `scripts/p6/`, `docs/run-cards/p6/`, `manifests/p6/`, `results/p6/`, `docs/hub/p6-m-schedule-topology/` under https://github.com/pageman/nanochat-filipino; Hub https://huggingface.co/pageman/nanochat-filipino-p6-m-schedule-topology (uploaded commit `5d3872b0`; topology children technical recreate). Run ID: `p6-20260824T155226Z-769f807a`. Manuscript version: v1.0.

1 Introduction

Pajo [1] measured equal-exposure Tagalog BPB across nanochat depths on WikiText-TL-39 (AsPredicted #306780). Pajo [2] asked the forward continual-pretraining question from an English parent (AsPredicted #306935). Pajo [3] asked the reverse from a fresh Tagalog parent (AsPredicted #307342). Pajo [4] locked source-content token share at $q_{\mathrm{TL}} = 0.50$ after a fresh Tagalog parent (AsPredicted #307591). Pajo [5] counted how often that frozen mix recurred on three unused initializations (AsPredicted #307836).

P6-M was designed after those released P1.1–P5 findings were known. It is therefore a prospectively preregistered **post-P5** schedule-topology mechanism study [6]. It is not an amendment, correction, rescue, replication, or independent confirmation of P4 or P5. It does not treat recurrence counts as primary. It holds the P4 token-share bundle fixed and varies only the filed block schedules.

The confirmatory question, locked before any P6 confirmatory BPB existed, is:

> After a newly trained seed-4 Tagalog parent passes P0-T, freeze d20 as C0 and continue it for one shared phase-two token budget on C1, C2, and four mixed topologies that share identical within-language streams and exact per-language quotas. Relative to M-fine, how do M-coarse, M-blocked, and M-rand classify on Tagalog and English validation BPB at $\delta = 0.01$?

This paper contributes: (i) one eligible seed-4 Tagalog parent frozen as C0; (ii) six sibling children under a fixed token budget; (iii) a sealed twelve-cell validation matrix before one M-fine-only secondary test; (iv) one Gate X release whose primary result is the six primary ΔTL/ΔEN classifications versus M-fine.

2 Related work

Cruz and Cheng constructed WikiText-TL-39 [7]. Sequential interference is classical [14, 15]. LLM continual-pretraining work often continues an English or multilingual base [12, 13, 11]. P6-M stays inside WikiText-family text, the nanochat depth dial, and BPB rather than CORE or chat [9]. Equal token budgets follow the family so stream identity is not confounded with compute [16, 17, 18]. WikiText established clean Wikipedia LM evaluation in English [10]. BPB remains comparable across vocabularies [19, 20]. Official P6 BPB is mean token NLL divided by $\ln 2$ times mean UTF-8 bytes per token, full split, $T = 2048$, BOS-best-fit. In-loop trainer BPB is not confirmatory. Preregistration reduces undisclosed flexibility [21, 22]. Curriculum learning motivates asking whether presentation order matters when exposure totals are locked [23]; surveys of continual LLM training likewise treat schedule and interference as open mechanism questions [11]. P6-M answers the block-order question only inside this one-seed BPB apparatus. WikiText-TL-39 is a Cruz–Cheng resource lineage; this study is not an extension of Cheng catastrophic-forgetting methods and does not assess downstream Filipino tasks [8].

3 Methods

3.1 Registration and authority

AsPredicted #307969 governs [6].

- PDF: `https://aspredicted.org/bk6m9d.pdf`
- PDF SHA-256:
  `769f807a00264a996b02c38b83ee2cf6c23c4981e36fb477dfc1959fa918a1e7`
- Prefiling addendum SHA-256:
  `df49664809ada69d23dcd2c799e75b30f0fdb9afd7aee12b071ac24ef2f81082`
- Unmodified gate-plan SHA-256:
  `d8a63608608c59d2c4d9882e5346625462056c331094942a8a01d496697a1c79`
- Topology-manifest SHA-256:
  `d1e7d5af7247a572e319ee003b5f4e3da5d1fb1592e5ed9ff6b22eeec15ea606`

Authority order: filed PDF $\gg$ SHA-bound addendum $\gg$ unmodified gate plan $\gg$ `LOCK.json`, manifests, and deterministic receipts $\gg$ this manuscript (v1.0). The study does not amend #306780, #306935, #307342, #307591, or #307836. ResearchBox #8918 remains FOR PEER REVIEW and is not made public by this paper. AsCollected #2541 Version 1 (`https://ascollected.org/XZ8_TI5`) records public-corpus provenance and is linked from the box. Hub weights were uploaded after Gate W with independent remote re-hash; topology children on Hub are a technical recreate (new SHAs).

3.2 Sources, tokenizer, and hosts

Tagalog train/val/test documents are the frozen WikiText-TL-39 P1.1-eligible splits. English train/val/test documents are the frozen WikiText-103 P2-eligible splits. The tokenizer is the carry-forward P4 pair: `tokenizer.pkl` `04436b854e0841025a3dd2b46baaeeea07a7ccc252e9f99a19171306f00bc5a8` and `token_bytes.pt` `a5dbc1c88f6292696108263072d77115718cc2d8357f7ad4859adfa517cc2132`. P1.1–P5 weights were not loaded as parents. Official GPU gates used NVIDIA CUDA A40. Official evaluation used `scripts/p6/evaluate_bpb.py` with formula-equivalent hash `a22a566b71ac9b768543e052e78576f6b079d5c88066f3ec2f8bfef600e17481`.

3.3 Parent, P0-T, and C0 freeze

Parent-initialization seed is exactly $s = 4$. After P0-T PASS, d20 was copied to an immutable C0 directory with `additional_train_tokens`=0. C0 SHA-256: `7fbd24de792aa4ee27d841866db2114e0fb45b1fdcaa4edcc9b24582220123c9`. Smoke and d8 checkpoints are not C0.

3.4 Children and fixed topology treatments

C1, C2, M-fine, M-coarse, M-blocked, and M-rand each continued the same frozen C0 for $N = 294$ steps, $B = 65{,}536$, $D_{\text{phase2}} = 19{,}267{,}584$ tokens, serial R$\to$S$\to$T1–T4, `load_optimizer=False`, child peak learning rate 0.3$\times$ the parent peak, warmup 14, terminal checkpoint only. Mixed arms share identical within-language streams and stored no-wrap schedules:

- **M-fine:** EN-first alternating blocks of 2,048 tokens (4,704 blocks per language).
- **M-coarse:** EN-first alternating blocks of 1,204,224 tokens (eight blocks per language).
- **M-blocked:** all Tagalog then all English (one block per language).
- **M-rand:** one precomputed shuffle of 4,704 EN and 4,704 TL blocks of length 2,048, shuffled once with Python `random.Random(42)` and hash-pinned before filing.

Each mixed arm uses exactly 9,633,792 Tagalog and 9,633,792 English source-content tokens. Mix-manifest SHA-256: `f203c615266bc8c33c358c1de397715791cae33536a9743c8a6bf8cd543cb107`.

3.5 Lockbox, analysis, and tests

Validation BPB was lockboxed until one Gate X. Gate U sealed twelve child cells (six conditions × two languages) with `test_access`=0. Policy A authorized exactly one M-fine-only secondary restricted-test event after U (Gate V). C1, C2, M-coarse, M-blocked, and M-rand were not tested. Primary contrasts use sealed validation cells; tests are secondary and excluded from topology classification. Equality outside $(-\delta, +\delta)$ at $\delta = 0.01$ defines directional classes. No composite score, mean, confidence interval, p-value, across-seed average, or P5-style recurrence count.

4 Results

P0-T status is **PASS** for seed 4. All six children completed with reload-verified terminal checkpoints. Table 1 reports the sealed twelve validation cells and descriptive C0 English. Table 2 reports the six primary contrasts versus M-fine. Table 3 reports contextual R_{TL} and A_{EN} for the four mixed topologies.

Table 1: Full-split validation BPB after terminal d20 checkpoints. Six-decimal reporting as filed. C0 English is descriptive and excluded from topology classification.

Arm	Tagalog val BPB	English val BPB
C0 (frozen parent)	—	2.683814
C1 extra Tagalog	0.564548	2.892551
C2 pure English	2.219915	1.269724
M-fine	1.256292	1.440084
M-coarse	1.214675	1.487389
M-blocked	1.458898	1.529111
M-rand	1.257379	1.441159

Table 2: Primary contrasts versus M-fine at $\delta = 0.01$. $\Delta = \mathrm{BPB}(M\text{-}\tau) - \mathrm{BPB}(\text{M-fine})$. Lower BPB is better.

τ	ΔTL	TL class	ΔEN	EN class
M-coarse	−0.041617	better than M-fine by δ	0.047305	worse than M-fine by δ
M-blocked	0.202606	worse than M-fine by δ	0.089027	worse than M-fine by δ
M-rand	0.001087	within δ	0.001075	within δ

Table 3: Contextual contrasts (descriptive secondary). $R_{\mathrm{TL}} = \mathrm{TL}(M\text{-}\tau) - \mathrm{TL}(\mathrm{C2})$; $A_{\mathrm{EN}} = \mathrm{EN}(M\text{-}\tau) - \mathrm{EN}(\mathrm{C1})$.

τ	R_{TL}	A_{EN}
M-fine	−0.963623	−1.452467
M-coarse	−1.005239	−1.405162
M-blocked	−0.761016	−1.363439
M-rand	−0.962536	−1.451392

Under this preregistered fixed-budget schedule comparison, M-rand was within δ of M-fine on both languages; M-blocked was worse than M-fine by more than δ on both languages; M-coarse was better than M-fine on Tagalog and worse on English, each by more than δ. This is a one-seed topology classification, not a confidence interval, not a population effect, and not a P5 recurrence count. P6-M does not amend #307591 or #307836.

One authorized M-fine-only secondary test event reported English test BPB 1.450423 and Tagalog test BPB 1.260475. Tests do not alter sealed contrasts and are not co-primary.

5 Discussion

With token share held fixed, block order was not inert in this apparatus: coarse interleaving and complete blocking moved validation BPB relative to fine EN-first alternation, while a once-shuffled random block sequence stayed within δ of M-fine. That pattern is a mechanism observation under a frozen P4-style quota bundle, not a claim that M-fine is optimal, not a general forgetting-mitigation result, and not a law across seeds. Contextual R_{TL} and A_{EN} place each mixed arm relative to C2 and C1; they are not a second topology success rule. WikiText-TL-39 remains a Cruz–Cheng resource lineage; P6-M does not extend Cheng catastrophic-forgetting protocols and does not claim downstream Filipino-task or all-Philippine-language benefit.

6 Limitations

One seed; one tokenizer; WikiText-family text only; no chat, SFT, CORE, FilBench, or larger-model transfer. Confirmatory GPU work used a rented NVIDIA A40. Raw test JSONL remains restricted. Hub weight release, if any, must ship C0+C1+C2+topology children together with tokenizer and meta; never overwrite P4/P5 Hub IDs. Document-revisit (P7-M), optimizer-state (P8-M), replay/protection, and tokenizer/transfer studies remain separate filings.

7 Deviations

No protocol stop. No break-glass. No additional confirmatory validation or test eval after Gate X. Gate S attempt 1 failed as a *preflight* block when the C2 English packed stream was missing on the pod path ($checkpoint\_sha256 = \texttt{null}$); after syncing the filed stream, one clean authorized C2 launch completed. M-rand wrote a valid terminal model checkpoint, then hit ENOSPC during optimizer save; the terminal model was hash- and reload-verified and accepted without retraining (`technical_accept=true`). Gate 0 had pinned a P5-rename unblind script hash that reads `c3_*` cells and emits recurrence counts; Gate U wrote topology-named cells, so Gate X executed a topology-contrast script under filed-analysis authority. The script-hash mismatch is documented in the unblinding event; the filed primary remains ΔTL/ΔEN versus M-fine. Frozen English train JSONL contains intra-split duplicate rows (28,472 rows / 28,232 unique); they were not dropped. Cross-split overlap was 0.

Availability

- AsPredicted #307969: https://aspredicted.org/bk6m9d.pdf
- Study record (GitHub):
  https://github.com/pageman/nanochat-filipino/tree/main/docs/papers/p6-m-schedule-topology
- Code and audit trees (GitHub):
  https://github.com/pageman/nanochat-filipino
  P6-only: `scripts/p6/`, `docs/papers/p6-m-schedule-topology/`, `docs/run-cards/p6/`, `manifests/p6/`, `docs/hub/p6-m-schedule-topology/`
- Weights (Hugging Face Hub):
  https://huggingface.co/pageman/nanochat-filipino-p6-m-schedule-topology
  (uploaded; commit `5d3872b0`; topology children technical recreate; never write onto P4/P5 Hub IDs).
- ResearchBox #8918: https://researchbox.org/8918 (FOR PEER REVIEW; not Make Public).
- AsCollected #2541 Version 1: https://ascollected.org/XZ8_TI5 (project `NANOCHAT-FILIPINO P6-M`; linked from box #8918).

Held-out test text is not redistributed. Host credentials are not published. GitHub holds protocol, receipts, sealed/released JSON, and paper; Hugging Face holds the weight bundle after independent remote re-hash.

Ethics

Public Wikipedia-derived corpora. No human-subjects experiment. ResearchBox #8918 remains passcode-protected for peer review until explicitly made public.

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A Selected hashes

- Pin / nanochat: `92d63d4e8bb4df75c3b71618f31ddde2378b2bcd`
- AsPredicted PDF: `769f807a00264a996b02c38b83ee2cf6c23c4981e36fb477dfc1959fa918a1e7`
- Gate plan: `d8a63608608c59d2c4d9882e5346625462056c331094942a8a01d496697a1c79`
- Addendum: `df49664809ada69d23dcd2c799e75b30f0fdb9afd7aee12b071ac24ef2f81082`
- Topology manifest: `d1e7d5af7247a572e319ee003b5f4e3da5d1fb1592e5ed9ff6b22eeec15ea606`
- Mix manifest: `f203c615266bc8c33c358c1de397715791cae33536a9743c8a6bf8cd543cb107`
- Evaluator (P6 path): `a22a566b71ac9b768543e052e78576f6b079d5c88066f3ec2f8bfef600e17481`
- C0: `7fbd24de792aa4ee27d841866db2114e0fb45b1fdcaa4edcc9b24582220123c9`
- C1 / C2: `6223c116779ce3128c1e4cae0f2e03744b3068169ad7bb88f75a5146671a99bd` / `04c06c195e513c7b30752637b80591c29e740879d91c35c323fb75fafca8747d`
- M-fine / M-coarse / M-blocked / M-rand: `a0139607a8fdf2772d4b8e722b449b6a8db04056a8dc38cd177708af8f15eeab` / `86588ee9f72ea4a85a821c2c882a363d55d3e50ccb325bdabaa0706e1e911dfe` / `5ec45216eb0a09b5bc04f04f4622d85f7d8f7ff8861fdea6de5487f7c18fa526` / `38580cd501c0d87a1a502397697f7c7e427134eb7cb64bdce2ba6be1a036f108`
- U seal: `23f9a94483532f5b7d51f4a27f77436cafd440f30c11fac5e0ea2dfa0aa095ae`
- Released contrasts: `d376643af19969f2ee127afcc4696de3a4821c8188fa6e7003d190d5f4031911`

B GitHub versus Hugging Face

GitHub `pageman/nanochat-filipino` is the study record: scripts, paper, lock, ledgers, run-card receipts, sealed/released JSON, and Hub *documentation*. Hugging Face `pageman/nanochat-filipino-p6-m-sc` is the weight deposit: C0+C1+C2+topology children plus tokenizer and evaluation JSON (uploaded; topology children technical recreate). Never a single topology child alone. Never write onto P1.1/P2/P3/P4/P5 Hub IDs. Raw test JSONL, passcodes, SSH keys, optimizer states, and HOST operator cards belong in neither public tree.

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